29 SYNTHETIC POLYMERS

Polymers are everywhere, ranging from naturally occurring polymers, for example proteins (polypeptides) and sugar (polysaccharides), which you probably have met before, to synthetic polymers used for plastic bags and clothing. Polymers are large molecules consisting of many repeating units. This chapter looks at two different ways that polymers can be made: addition polymerisation for polymers such as poly(ethene) and PVC, and condensation polymerisation for polymers such as polyester.





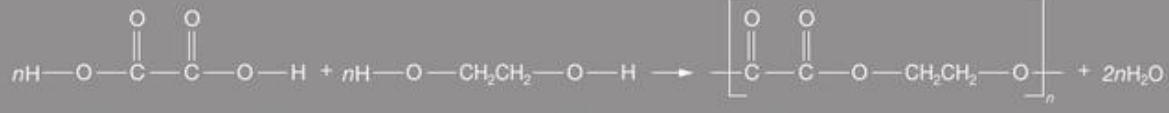
▲ Figure 29.1 Synthetic polymers are extremely useful for making a variety of products, from the coating for electrical wires to plastic bottles and intelligent label chips.

LEARNING OBJECTIVES

- Know that an addition polymer is formed by joining up many small molecules called monomers
- Understand how to draw the repeat unit of an addition polymer, including poly(ethene), poly(propene), poly(chloroethene) and poly(tetrafluoroethene)
- Understand how to deduce the structure of a monomer from the repeat unit of an addition polymer and vice versa
- Explain problems in the disposal of addition polymers, including:
  - their inertness and inability to biodegrade
  - the production of toxic gases when they are burned

CHEMISTRY ONLY

- Know that condensation polymerisation, in which a dicarboxylic acid reacts with a diol, produces a polyester and water.
- Understand how to write the structural and displayed formula of a polyester, showing the repeat unit, given the formulae of the monomers from which it is formed, including the reaction of ethanedioic acid and ethanediol:

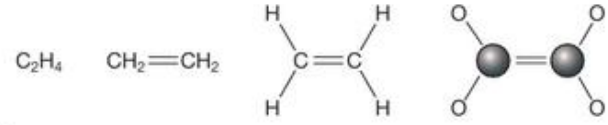


- Know that some polyesters, known as biopolyesters, are biodegradable.

ADDITION POLYMERISATION

THE POLYMERISATION OF ETHENE

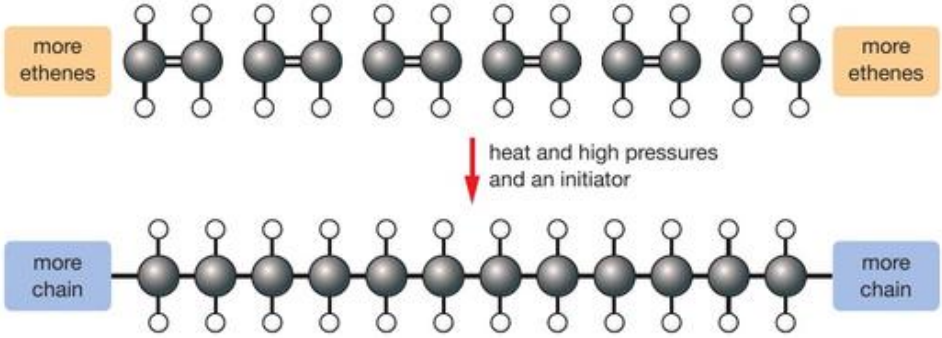
Ethene is the smallest alkene and can be produced by cracking. It is the smallest hydrocarbon containing a carbon–carbon double bond. Figure 29.2 shows different ways of representing an ethene molecule.



▲ Figure 29.2 Ethene

Under the right conditions, molecules containing carbon–carbon double bonds can join together to produce very long chains. Part of the double bond is broken to become a single bond, and the electrons in it are used to join to neighbouring molecules.

Polymerisation *is the joining up of lots of little molecules* (**monomers**) *to make one big molecule* (**polymer**). In the case of ethene, lots of ethene molecules join together to make *poly(ethene)*, which is more usually called *polythene*. *Molecules simply add onto each other without anything else being formed*. This is called **addition polymerisation**.



▲ Figure 29.3 The polymerisation of ethene

KEY POINT

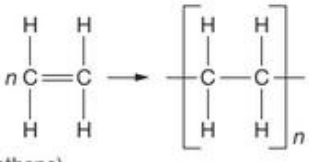
An *initiator* is used to start the process. You mustn't call it a catalyst because it is consumed in the reaction.

EXTENSION WORK

People occasionally wonder what happens at the ends of the chains. They don't end tidily! Bits of the initiator are bonded on at either end. You don't need to worry about that for International GCSE.

The chain length can vary from about 4000 to 40 000 carbon atoms.

For normal purposes, the polymerisation reaction is written using displayed formulae.

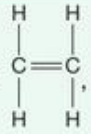
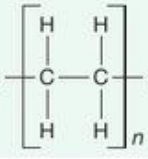
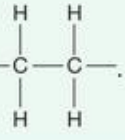


▲ Figure 29.4 Formation of poly(ethene)

KEY POINT

In this structure for poly(ethene) *n* represents a large but variable number. It simply means that the structure in the brackets (called the *repeat unit*) repeats itself many times in the molecule. When representing a polymer, the bonds on the two sides of the repeat unit should extend outside the square brackets to show that the carbon atoms are joined to the next ones in a chain, as shown in Figure 29.4.

HINT

Make sure you can distinguish between the different terms monomer, polymer and repeat unit. In the equation in Figure 29.4, the monomer is  , the polymer is  and the repeat unit is  .

USES FOR POLY(ETHENE)

Poly(ethene) comes in two types: low-density poly(ethene) (LDPE) and high-density poly(ethene) (HDPE). Low-density poly(ethene) is mainly used as a thin film to make polythene bags. It is very flexible and not very strong.

KEY POINT

Although the polymer is called *poly(ethene)* it is actually an *alkane*. The polymer is saturated and will not decolourise bromine water.

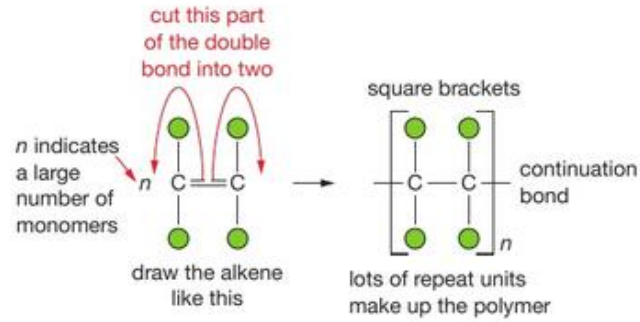


▲ Figure 29.5 Plastic bags made from poly(ethene)

High-density poly(ethene) is used where greater strength and rigidity are needed, for example to make plastic bottles such as milk bottles. If you can find a recycling symbol with the letters HDPE next to it, then the bottle is made of high-density poly(ethene). If it has some other letters there, then it is made of a different polymer.

HOW TO DEDUCE THE POLYMERISATION REACTION FOR ANY ALKENE

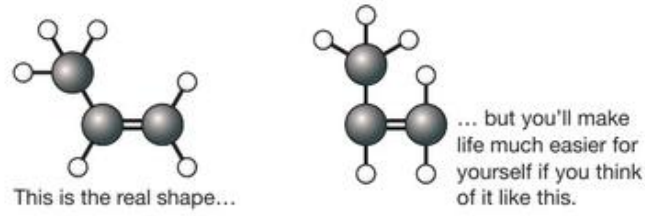
Figure 29.6 shows how to work out a polymerisation reaction starting from any alkene. The key thing is to always draw the alkene as shown with just the C=C in the middle.



▲ Figure 29.6 How to work out a polymerisation reaction

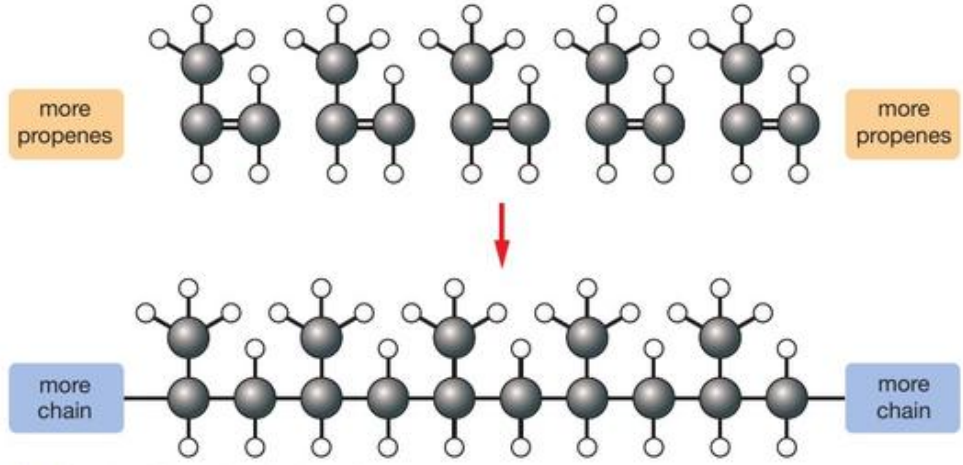
THE POLYMERISATION OF PROPENE

Propene is another alkene, this time with three carbon atoms in each molecule. Its formula is normally written as $CH_3CH=CH_2$. Think of it as a modified ethene molecule, with a CH_3 group attached in place of one of the hydrogen atoms.



▲ Figure 29.7 Propene

When propene is polymerised you obtain *poly(propene)*. This used to be called *polypropylene*.

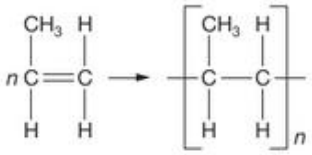


▲ Figure 29.8 The polymerisation of propene

HINT

When you draw an addition polymerisation reaction never change anything except opening out the double bond and drawing the continuation bonds on each side.

Write this as:



▲ Figure 29.9 Formation of poly(propene)

USES OF POLY(PROPENE)

Poly(propene) is somewhat stronger than poly(ethene). It is used to make ropes and crates (among many other things). If an item has a recycling mark with PP inside it or near it, it is made of poly(propene).



▲ Figure 29.10 PP recycling symbol



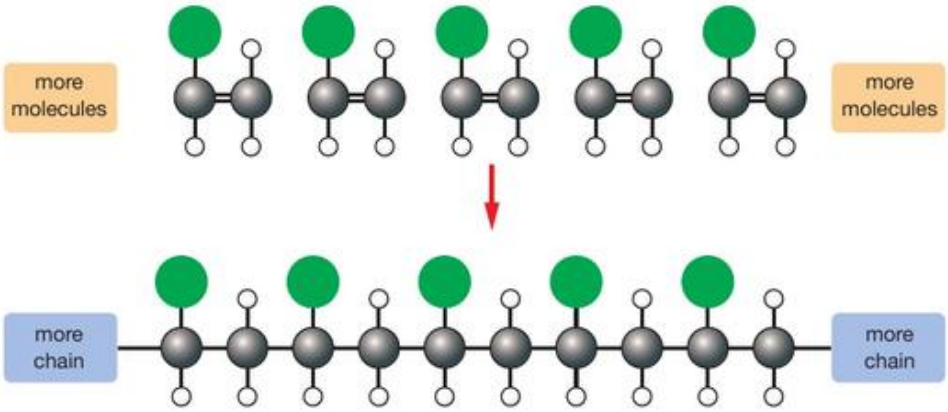
▲ Figure 29.11 Poly(propene) is used to make crates . . .



▲ Figure 29.12 . . . and ropes.

THE POLYMERISATION OF CHLOROETHENE

Chloroethene is a molecule in which one of the hydrogen atoms in ethene is replaced by a chlorine. Its formula is CH_2=CHCl. In the past, it was called vinyl chloride. Polymerising chloroethene gives you poly(chloroethene). This is usually known by its old name, polyvinylchloride or PVC.

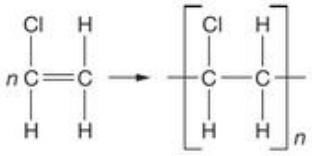


▲ Figure 29.13 The polymerisation of chloroethene

Write this as:

HINT

When you draw this, it isn't important whether you put the chlorine atom on the left-hand carbon atom or the right-hand one.



▲ Figure 29.14 Formation of poly(chloroethene)

USES OF POLY(CHLOROETHENE)

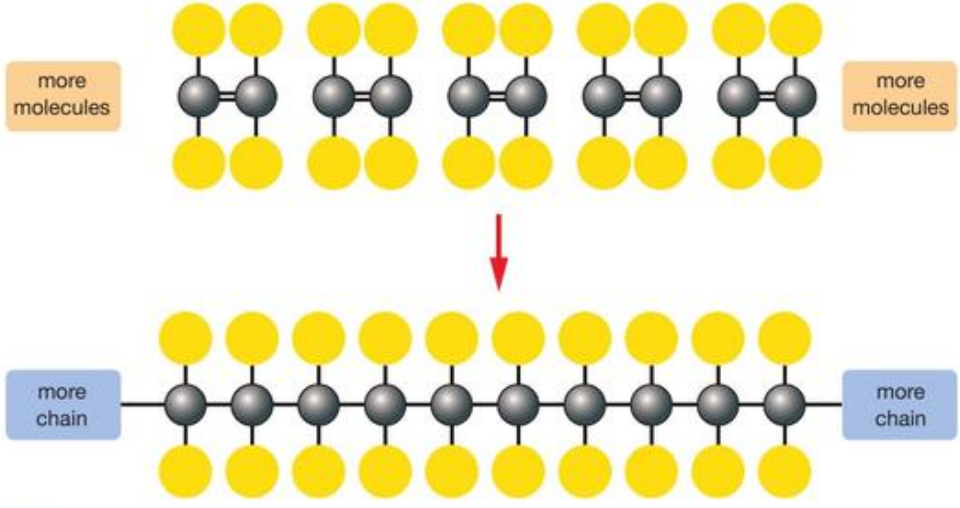
Poly(chloroethene), known as PVC, has a lot of uses. It is quite strong and rigid, and so can be used for water pipes or replacement windows. It can also be made flexible by adding *plasticisers*. This makes it useful for sheet floor coverings, and even clothing. These polymers don't conduct electricity, therefore PVC can be used for electrical insulation.



▲ Figure 29.15 PVC has a variety of uses.

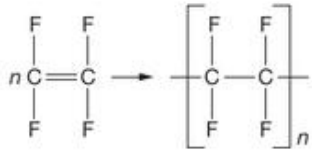
THE POLYMERISATION OF TETRAFLUOROETHENE

Tetrafluoroethene is another molecule derived from ethene in which all four hydrogen atoms are replaced by fluorine. Its formula is $CF_2=CF_2$. Polymerising tetrafluoroethene gives you poly(tetrafluoroethene) or PTFE, more commonly known as Teflon®.



▲ Figure 29.16 The polymerisation of tetrafluoroethene

Write this as:



▲ Figure 29.17 Formation of poly(tetrafluoroethene)

KEY POINT

If you look at this reaction, you can see it is exactly the same as the polymerisation of ethene except that each H has been replaced by an F.

USES OF POLY(TETRAFLUOROETHENE)

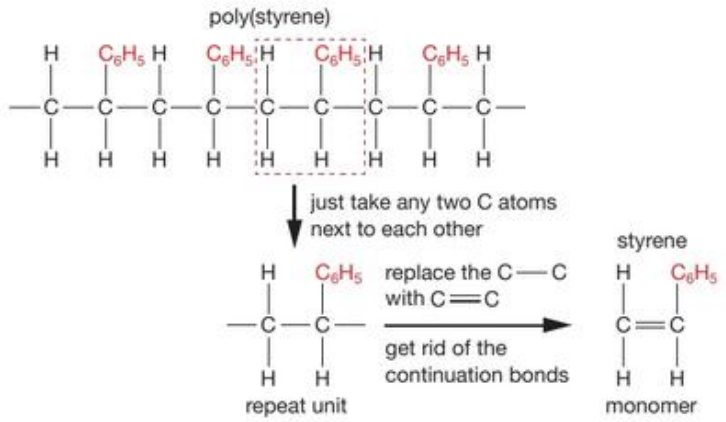
Poly(tetrafluoroethene) is often used as a non-stick coating for pots and pans. It is very unreactive due to the strong carbon–fluorine bonds, and can be found lining containers for corrosive chemicals.



▲ Figure 29.18 PTFE is used for non-stick coatings.

WORKING OUT THE MONOMER FOR A GIVEN ADDITION POLYMER

In an exam you may find that you are given the structure of a polymer and asked to deduce what the repeat unit is or what monomer it was made from. Figure 29.19 shows you how to do this.



▲ Figure 29.19 How to work out the repeat unit and monomer for poly(styrene)

The C_6H_5 group is complicated, but we don't need to worry about that when we write the structure of the monomer; work from the structure you are given. First find the repeat unit, which can be done simply by taking any two adjacent carbon atoms in the main polymer chain. To derive the monomer, a double bond needs to be put back between the two middle carbon atoms of the repeat unit.

DISPOSAL OF ADDITION POLYMERS

Recycling of plastics is important, not just to save raw materials, but because it takes a very long time for addition polymers like poly(ethene) and poly(chloroethene) to break down in the environment. They contain strong covalent bonds, making them essentially *inert* at ordinary temperatures. They are **non-biodegradable**, meaning *they cannot be broken down by bacteria in the environment*.

One solution to the problem of their disposal is to *bury them in landfill sites*, which are basically just big holes in the ground, where they will remain unchanged for thousands of years.



▲ Figure 29.20 Plastics (and other rubbish) can be disposed of in landfill sites.

Some countries, including Denmark and Japan, *incinerate (burn) plastics* in order to tackle the problem of disposal. This releases a lot of heat energy, which can be used to generate electricity. However, there are problems associated with this. For example, carbon dioxide is produced, which

contributes to **global warming**. Toxic gases are also released, including carbon monoxide and hydrogen chloride.

There are advantages and disadvantages for using either method to dispose of plastics.

Disposal method	Advantages	Disadvantages
Landfill	No greenhouse gases or toxic gases produced from plastics Cheap	Ugly, smelly and noisy; no one wants to live next to a landfill site Uses large areas of land The waste will be there for thousands of years
Incineration	Requires little space Can produce heat for local homes/offices and/or produce electricity	It is expensive to build and maintain the plant Produces greenhouse gases Releases toxic gases The ash produced must still be disposed of in landfill sites

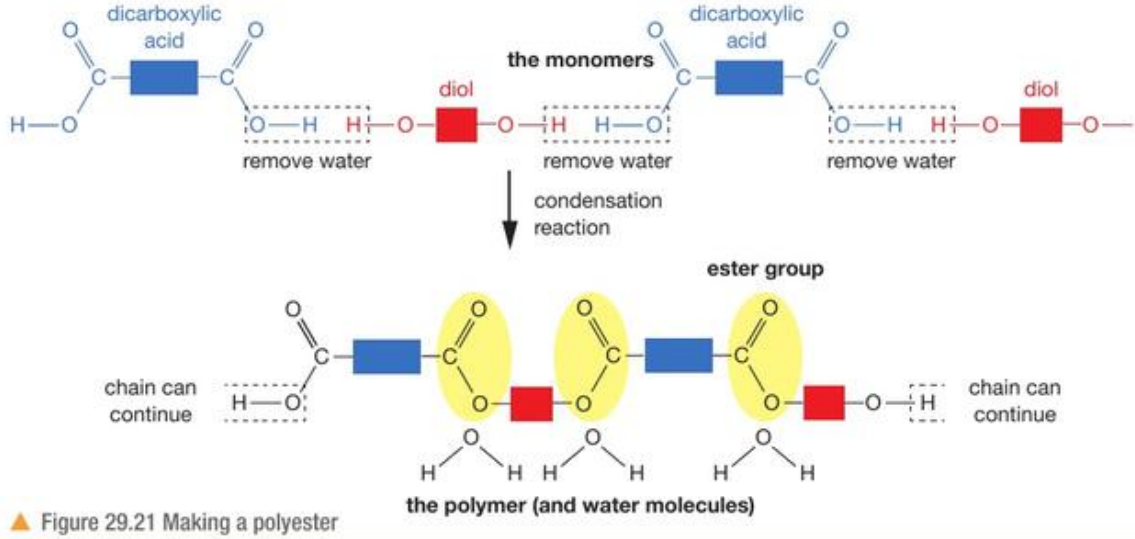
CHEMISTRY ONLY

CONDENSATION POLYMERISATION

MAKING A POLYESTER

When a lot of ethene molecules combine to make a chain, the double bonds break and the monomer molecules just add onto each other to make a polymer. *Nothing is lost*. That is why it is called *addition polymerisation*.

Making a polyester is different. Instead of one type of monomer, you often have two, joining together alternately. Each time two monomers combine, a small molecule such as water or hydrogen chloride is lost. The elimination of water gives the name *condensation* to this type of polymerisation.



▲ Figure 29.21 Making a polyester

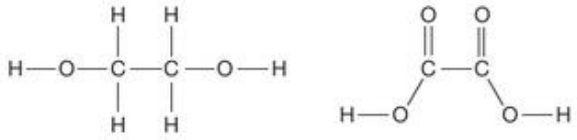
THE MONOMERS

One of the monomers is a *diol*, an alcohol molecule with one –OH at each end. For example, ethane-1,2-diol has the structural formula CH_2OHCH_2OH. The prefix 'eth' indicates the presence of two carbon atoms in the molecule. The code 'di' in the name shows that there are two –OH groups present. The number '1,2' tells you the –OH groups are attached to carbons number 1 and 2, one on each carbon. Propane-1,3-diol, which has one more carbon in the carbon chain, has the structural formula $CH_2OHCH_2CH_2OH$.

KEY POINT

It is not a mistake, there is an extra 'e' in each of these names. We talk about propan-1-ol but propan*e*-1,3-diol.

The second type of monomer is a *dicarboxylic acid*, a carboxylic molecule with one –COOH at each end. For example, ethanedioic acid has the structural formula HOOCCOOH. A more complicated example of a dicarboxylic acid is hexane-1,6-dioic acid, with the formula $HOOCCH_2CH_2CH_2CH_2COOH$, where 'hex' indicates six carbon atoms in the main chain of the molecule and '1,6-dioic acid' means there are two carboxylic acid functional groups at either end of the molecule, on carbons 1 and 6.



ethane-1,2-diol ethanedoic acid

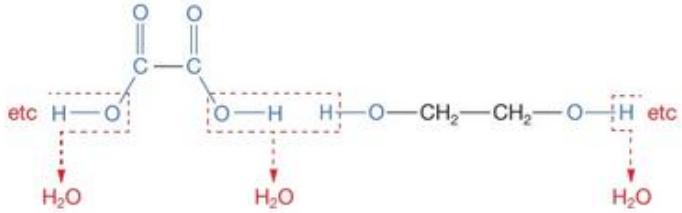
▲ Figure 29.22 Displayed formulae for ethane-1,2-diol and ethanedoic acid, monomers for a condensation polymerisation reaction to form a polyester.

FORMING A POLYESTER

HINT

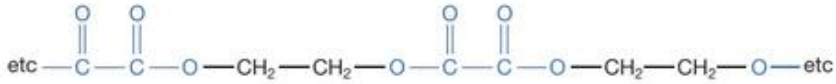
It does not matter whether you take the OH from the dicarboxylic acid and the H from the alcohol or the other way around – the polymer formed will still be the same.

Under the right conditions, diol and dicarboxylic molecules can join together with the loss of a molecule of water each time a new bond is formed.

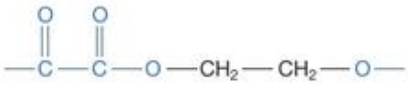


▲ Figure 29.23 Joining the monomers together...

Both monomers react at each end, joining together to form a long-chain polymer held together by a large number of ester groups, as shown in Figure 29.24.

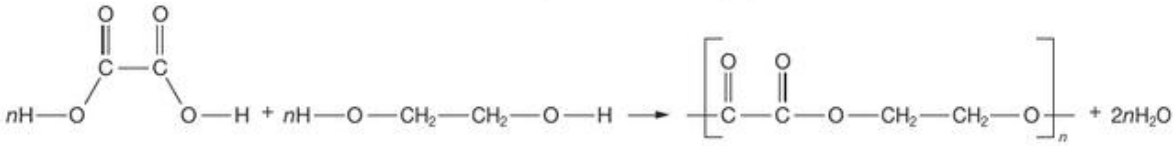


▲ Figure 29.24 ... to make a chain.



▲ Figure 29.25 Repeat unit of a polyester made from ethane-1,2-diol and ethanedoic acid.

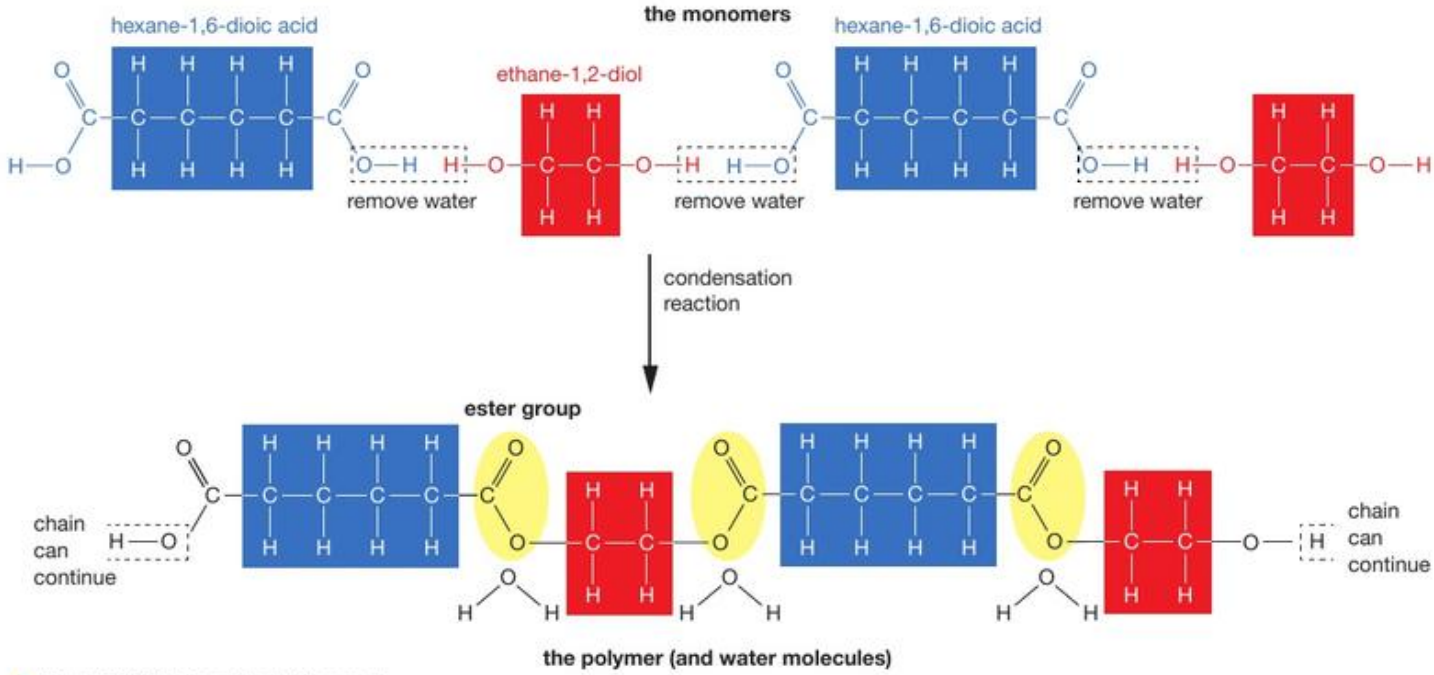
The balanced chemical equation for this polymerisation is:



EXTENSION WORK

The equation we have shown balances with $2n$ water molecules but in reality there will only be $(2n - 1)$ water molecules formed because there will still be one –OH and one –COOH left at the ends of the polyester molecule.

We have already seen one *block diagram* at the beginning of this section which shows the general condensation polymerisation reaction (Figure 29.21). A block diagram is a useful way of learning this because you can use the box to represent any hydrocarbon chain. For example, if we put in some carbons and hydrogens to make the reaction between hexane-1,6-dioic acid and ethane-1,2-diol, we obtain:



▲ Figure 29.26 Using a block diagram to work out the structure of any polyester.

HINT

This is how you should approach learning this. Think of one of the monomers as being a red rectangle with an –OH group at either end. Think of the other one as being a blue rectangle (or a different shape) with a –COOH group at either end. Draw those groups as displayed formulae. Then line them up so that you can remove the water between them. Having removed the water, slot the rest of the molecules together as shown in Figure 29.26. It looks difficult because it is unfamiliar. Practise doing it a few times until it becomes familiar.

BIODEGRADABLE POLYESTERS



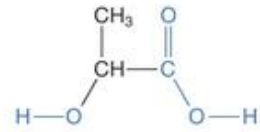
▲ Figure 29.28 A biodegradable plastic bag

DID YOU KNOW?

Lactic acid, more properly known as 2-hydroxypropanoic acid ($CH_3CHOHCOOH$), has both an alcohol and a carboxylic acid group in the same molecule so it can polymerise with itself (self-polymerise).

As with addition polymers there are environmental issues with the disposal of condensation polymers. These are more reactive because of the ester linkage and will all break down eventually, although this could take hundreds of years. Scientists have developed some *biodegradable polyesters* which break down much more quickly. These are called **biopolyesters**.

Biodegradable polymers can be made from lactic acid. Lactic acid can be obtained from corn starch, and when it undergoes polymerisation it forms a biodegradable polyester (polylactic acid, or PLA), which can be used for making biodegradable plastic bags or in surgery for internal stitches.



▲ Figure 29.27 Lactic acid monomer for making polylactic acid (PLA)

EXTENSION WORK

Amino acids, a group of compounds containing an *amine* group ($–NH_2$) and a *carboxylic acid* group (–COOH), can *self-polymerise* in condensation polymerisation reactions. The amine group reacts very similarly to the –OH group in alcohols, and it connects with a carboxylic acid to form an amide bond. When amino acids join together, they form polypeptides, which make up the primary structure of a protein.

END OF CHEMISTRY ONLY

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING 6

1 Ethene, C_2H_4, is an important molecule used in industry. It is an *unsaturated hydrocarbon*.

a Explain the terms *unsaturated* and *hydrocarbon*.

SKILLS PROBLEM SOLVING 9

b Ethene burns easily and releases a large amount of heat during combustion. Write a balanced chemical equation for the complete combustion of ethene.

SKILLS REASONING 7

c Explain why in industry ethene is considered too valuable to be used as a fuel.

SKILLS CRITICAL THINKING 6

2 Propene, C_3H_6, can be polymerised to make poly(propene).

a Explain the term *polymerisation*.

SKILLS INTERPRETATION 7

b Draw a displayed formula for propene.

8

c Draw a diagram to represent the structure of a poly(propene) chain showing three repeat units.

SKILLS CRITICAL THINKING 6

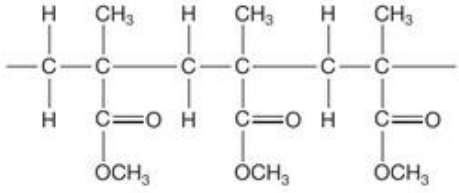
d The formation of poly(propene) is an example of *addition* polymerisation. Explain what is meant by the word *addition*.

SKILLS PROBLEM SOLVING 9

e Styrene has the formula $C_6H_5CH{=}CH_2$. Write a balanced chemical equation to show what happens when styrene is polymerised to make polystyrene. Your equation should clearly show the structure of the polystyrene. (Show the C_6H_5 group as a whole.)

SKILLS INTERPRETATION 8

f A small part of a Perspex molecule looks like this:



Draw the structure of the monomer from which Perspex is made.

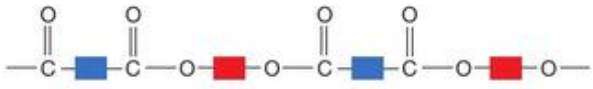
SKILLS DECISION MAKING 7

g Many addition polymers, for example those used for making plastic bags, are not biodegradable. Used plastic bags can be either buried underground in landfill sites or incinerated. State which method of disposal you think is better and give two reasons for your choice.

CHEMISTRY ONLY

SKILLS ANALYSIS

3 A polymer has the structure shown below:



a Identify the type of the polymer.

SKILLS INTERPRETATION

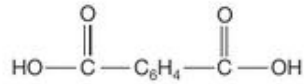
b Draw the structures of the monomers that are used to make this polymer.

SKILLS INTERPRETATION 8

4 Polyesters such as Terylene (for clothes) or PET (commonly used to make drinks bottles) are made by condensation polymerisation from ethane-1,2-diol and terephthalic acid (properly known as benzene-1,4-dicarboxylic acid). The structures of these are:

Ethane-1,2-diol: $HO–CH_2CH_2–OH$

Terephthalic acid:



a Draw a chain of the polymer with two repeat units.

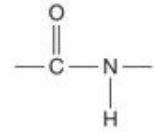
SKILLS PROBLEM SOLVING 9

b Write a balanced chemical equation for the formation of PET.

(The structure of the C_6H_4 group is complicated, and you can write it simply as C_6H_4. You can draw this as a block diagram if you wish, but it is more satisfying (and no more difficult in this case) to draw the structure properly.)

SKILLS CRITICAL THINKING 7

5 *(This question contains new material and is designed to look difficult. It is actually not too difficult as long as you understand about polyester.)* Nylon-6,6 is made by a condensation polymerisation of the monomers 1,6-diaminohexane ($H_2NCH_2CH_2CH_2CH_2CH_2CH_2NH_2$) and hexane-1,6-dioic acid. The amine group $–NH_2$ reacts in a very similar manner to the –OH group in an alcohol during condensation polymerisation with carboxylic acids, forming amide functional groups which join the monomers together into a polymer chain.



a i Explain what is meant by *condensation polymerisation* and how it differs from *addition polymerisation*.

SKILLS INTERPRETATION

ii Using a block diagram, draw one repeat unit for nylon-6,6.

SKILLS ANALYSIS, REASONING 8

b Nylon-6,10 is made from 1,6-diaminohexane and a longer chain acid, decanedioic acid, containing a total of 10 carbon atoms: $HOOCCH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2COOH$.

i How will a chain of nylon-6,10 differ from one of nylon-6,6? Refer to the diagram you drew in **a ii**.

ii In what way(s) will the two chains be the same? Again, refer to the diagram you drew in **a ii**.

END OF CHEMISTRY ONLY